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Branching angles in the modulation of plant architecture: Molecular mechanisms, dynamic regulation, and evolution

Chen Yun[1,2,4], Wanzhuang Ma[3,4], Jun Feng[3] and Lanxin Li[1,2,*]

[1]Frontiers Science Center for Molecular Design Breeding (MOE), China Agricultural University, Beijing, China

[2]Beijing Key Laboratory of Development and Quality Control of Ornamental Crops, Department of Ornamental Horticulture, College of Horticulture, China Agricultural University, Beijing, China

[3]College of Biological Science and Technology, Taiyuan Normal University, Jinzhong, China

[4]These authors contributed equally to this article.

***Correspondence: Lanxin Li (lanxin.li@cau.edu.cn)**

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ABSTRACT

Plants develop branches to expand areas for assimilation and reproduction. Branching angles coordinate with branching types, creating diverse plant shapes that are adapted to various environments. Two types of branching angle—the angle between shoots and the angle in relation to gravity or the gravitropic set-point angle (GSA) along shoots—determine the spacing between shoots and the shape of the aboveground plant parts. However, it remains unclear how these branching angles are modulated throughout shoot development and how they interact with other factors that contribute to plant architecture. In this review, we systematically focus on the molecular mechanisms that regulate branching angles across various species, including gravitropism, anti-gravitropic offset, phototropism, and other regulatory factors, which collectively highlight comprehensive mechanisms centered on auxin. We also discuss the dynamics of branching angles during development and their relationships with branching number, stress resistance, and crop yield. Finally, we provide an evolutionary perspective on the conserved role of auxin in the regulation of branching angles.

Key words: branching angles, plant architecture, auxin, tropisms, anti-gravitropic offset, evolution

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PLANT ARCHITECTURE IS DETERMINED BY BRANCHING ANGLES

Plants generate iterated growth units by branching, forming their body plan post-embryonically. The shape of a plant is predominantly influenced by two types of angles that involve the primary branches and their dynamic changes. The first type is the branching angle, the angle between a primary branch and the main stem (Figure 1A and 1B) or between two dichotomous branches in early-diverging plants (Figure 1C). These branching angles are initially established through interactions among shoot meristems. The second type of angle pertains to the direction opposite to gravity, the supplementary angle of the gravitropic set-point angle (GSA). The angles at the bases or growing tips of primary branches are termed the base or tip angles, respectively. The base angle aligns with the branching angle when the main stem is straight (Figure 1A, 1B, 1D, and 1E). For grass-like plants such as rice, which lack a distinct main stem, a similar base angle is referred to as the tiller angle and represents the angle of the tillers at maximum inclination relative to the direction opposite to gravity (Figure 1F). The tip angle describes the angle formed between the extension line of the growing tip and the direction opposite to gravity (Figure 1G–1I). In general, the base and tip angles outline the plant's overall structure. When the base and tip angles of the primary branches are acute, the plant exhibits an erect or pillar-like form (Figure 2A). By contrast, a nearly right or obtuse base angle, or an increasing tip angle, leads to a prostrate growth habit (Figure 2B, 2D, 2E, and 2G). A range of wide and gradient base and tip angles along the main stem or throughout the overall architecture of plants is essential for creating a spherical (Figure 2C–2F and 2I) or a conical shape (Figure 2H).

More intricate changes occur along the branch, causing it to grow gradually upward or downward in relation to gravity. The dynamics of the tip angle correspond with shoot branching. A downward-growing branch is not sufficient (Figure 2D) but is required for epitony branching, in which the upper axillary branches grow faster (Figure 2G); this is often observed in fruit

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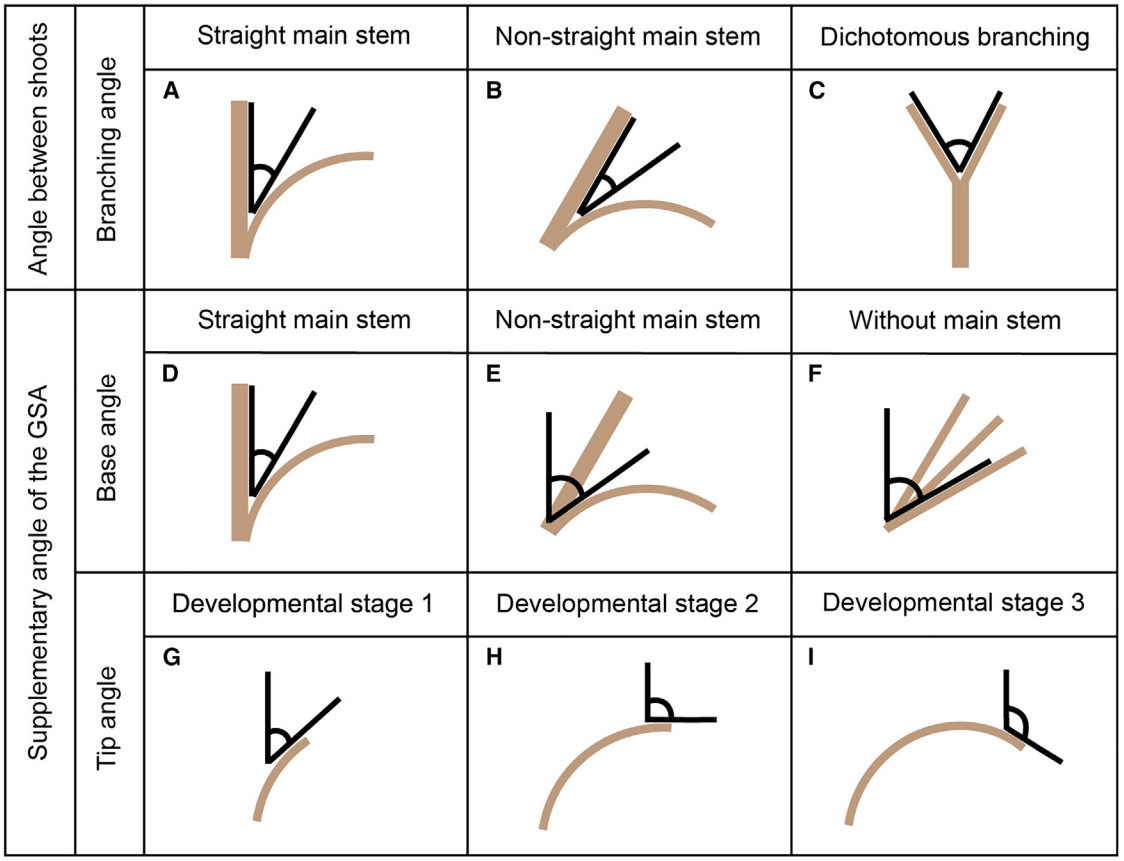

Figure 1. Types of branching angles.
Branching angles include the angles between shoots **(A–C)**, the base angle **(D–F)**, and the tip angle **(G–I)**. The branching angle is equal to the base angle when the main stem is straight **(A and D; B and E)**. The branching angle refers to the angle between two dichotomous branches in early-diverging plants **(C)**. In grasses without a main stem, such as rice, the base angle often corresponds to the tiller angle **(F)**. The tip angle changes throughout branch development **(G–I)**. The thick shoot in brown indicates the main stem, and the thin shoots represent primary branches.

trees and landscape trees selected for ease of harvest or aesthetic value (Barthélémy and Caraglio, 2007). By contrast, an upward-growing branch is required for hypotony branching, which is characterized by accelerated growth of the lower axillary branches. The direction of growth in amphitony branching remains consistent as branches develop symmetrically around the main stem. A combination of hypotony and amphitony branching can produce a crown-like structure (Barthélémy and Caraglio, 2007), as seen in herbaceous plants such as *Bassia scoparia* and the groundcover *Chrysanthemum morifolium* (Figure 2C), shrubs like *Pittosporum tenuifolium* 'Golf Ball' (Figure 2F), and trees such as *Araucaria araucana* and *Salix matsudana* 'Umbraculifera' (Figure 2H and 2I). Therefore, the dynamics of the tip angle, along with shoot branching, play a crucial role in shaping the aboveground architecture of plants.

Overall, branching angles shape plant shoot architecture, thereby directly affecting the side of the plant that receives light, the side that interacts with neighboring plants in a population, and the area where plant branches contact one another. These serve as the basis for the crucial role of branching angles in determining plant architecture, photosynthetic efficiency in dense plantings, and agronomic traits, including yield and stress resistance (Sakamoto et al., 2006; Burgess et al., 2017; Sun et al., 2019; Wang et al., 2022; Basu and Parida, 2023). Understanding the mechanisms that regulate branching angles in model plants, crops, and woody plants provides valuable insights into the conserved and specialized regulatory networks that shape plant architecture. Furthermore, such an understanding will offer a genetic toolbox for molecular breeding in agriculture and horticulture.

THE REGULATORY MECHANISMS OF BRANCHING ANGLES

The angle of a growing branch at any position is in principle formed by differential cell expansion or division between upper and lower cells, mainly regulated by tropisms, hormones, transcription factors, and response to environmental changes (Li et al., 2022). This chapter discusses how gravitropism, anti-gravitropic offset (AGO), light signaling, and other regulators modulate branching angles.

Gravitropism: Various responses to ever-present gravity

Perception: Amyloplast formation, sedimentation, and amyloplast-independent sensing

The best understood mechanism for regulation of shoot branching angles is gravitropism. Gravity is sensed in specialized cells

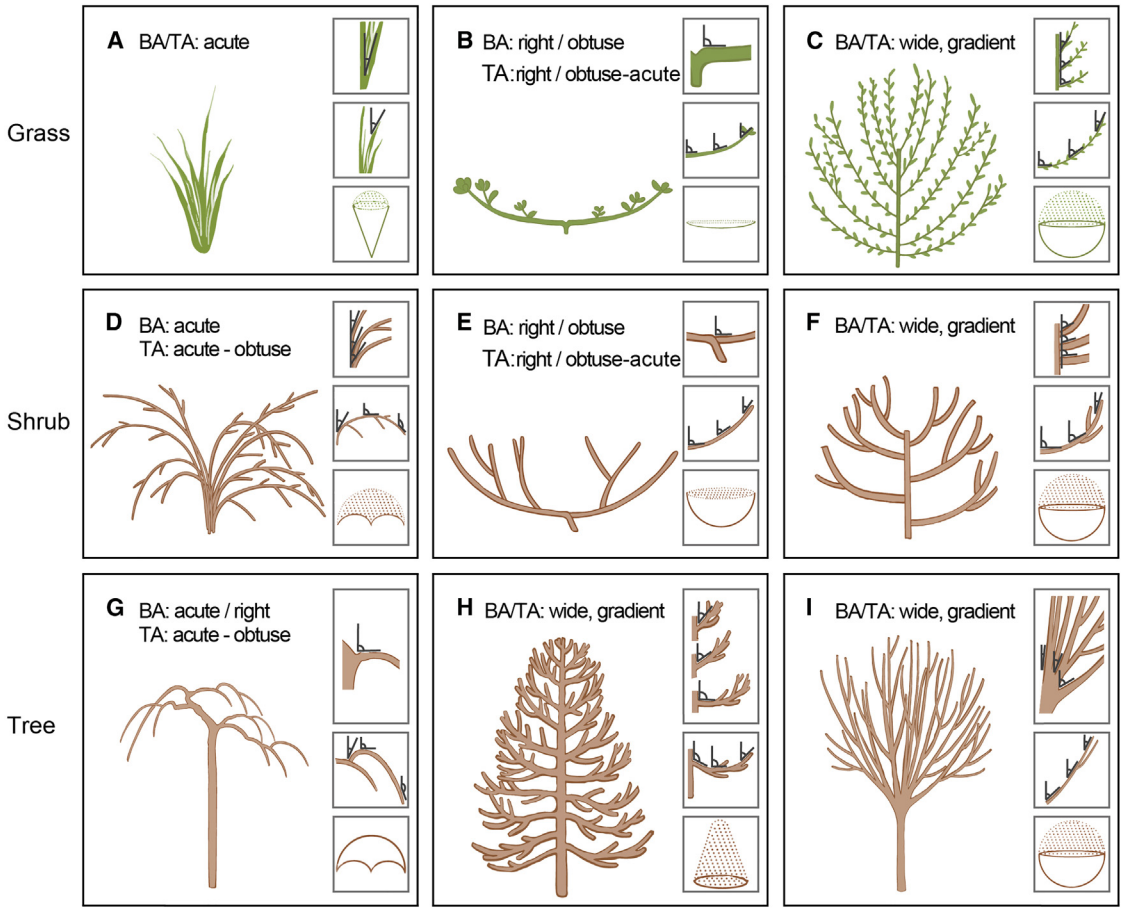

Figure 2. The base and tip angles shape plant architecture.

The illustrated plant architectures include those of grasses **(A–C)**, shrubs **(D–F)**, and trees **(G–I)**. Each subfigure contains three small vertical panels. The upper panel shows the base angle (BA); the middle panel shows the tip angle (TA) during branch development; and the lower panel shows an abstract three-dimensional (3D) model of the plant architecture surface. In this 3D model, the dots represent the surface composed of branch tips, and the solid lines indicate the surface formed by the branches. The shape of the 3D model is determined by BA, TA, and branch length. Similar architectures of the 3D models can be found in **(B)** and **(E)**; **(C)**, **(F)**, and **(I)**; and **(D)** and **(G)**.**(A)** *Lolium perenne* grows upright with an acute BA and TA.

(B and E) *Portulaca oleracea* **(B)** and *Juniperus procumbens* **(E)** display a prostrate phenotype, characterized by a right or obtuse BA and a dynamic TA that transitions from right or obtuse in the early stages to acute in the later stages.

(C, F, H, and I) *Bassia scoparia* or the groundcover *Chrysanthemum morifolium* **(C)**, *Pittosporum tenuifolium* 'Golf Ball' **(F)**, and *Salix matsudana* 'Umbraculifera' **(I)** exhibit spherical forms, whereas *Araucaria araucana* **(H)** displays a conical shape. All these species exhibit a BA and TA that span a wide and gradient range.

(D and G) Both *Jasminum nudiflorum* **(D)** and *Styphnolobium japonicum* 'Pendula' **(G)** feature downward-growing branches with a dynamic TA that starts acute and becomes obtuse over time. The BA of *J. nudiflorum* is acute, whereas that of *S. japonicum* 'Pendula' can be either acute or right.

called statocytes, which contain starch granules that physically sediment in the direction of gravity. In the shoots of seed plants, statocytes are distributed along the shoots, including the inflorescence stems of eudicots as well as the coleoptiles and leaf-sheath pulvinus of monocots (Kawamoto and Morita, 2022). This broad distribution facilitates gravity perception, which may underlie the dynamic regulation of branching angles throughout shoot development. By contrast, statocytes in the roots of seed plants are primarily confined to the columella cells at the root tips, allowing for rapid gravitropic responses (Zhang et al., 2019b). The nature and regulation of amyloplast sedimentation in shoots also differ from those in roots. Studies in *Arabidopsis* have shown that amyloplasts in the endodermal cells of primary inflorescence stems exhibit thylakoid membrane structures that resemble those of chloroplasts (Morita et al., 2002). This suggests that amyloplasts in the shoot endodermis are likely specialized chloroplasts that integrate the processes of photosynthesis and gravity perception, unlike the amyloplasts found in roots (Morita, 2010; Kawamoto and Morita, 2022). In addition, amyloplast sedimentation in shoots is regulated by the enwrapping central vacuoles, which differ from the fragmented small vacuoles in root columella cells (Saito et al., 2005). It has been observed that amyloplast sedimentation in *Arabidopsis* shoots can occur in as little as 3 min (Saito et al., 2005), but it remains unclear whether woody branches exhibit a similarly rapid response.

Because starch granules perceive gravity at the onset of gravitropism, mutations in starch biosynthesis genes, such as *PHOSPHOGLUCOMUTASE* (*PGM*) and *ADP-GLUCOSE*

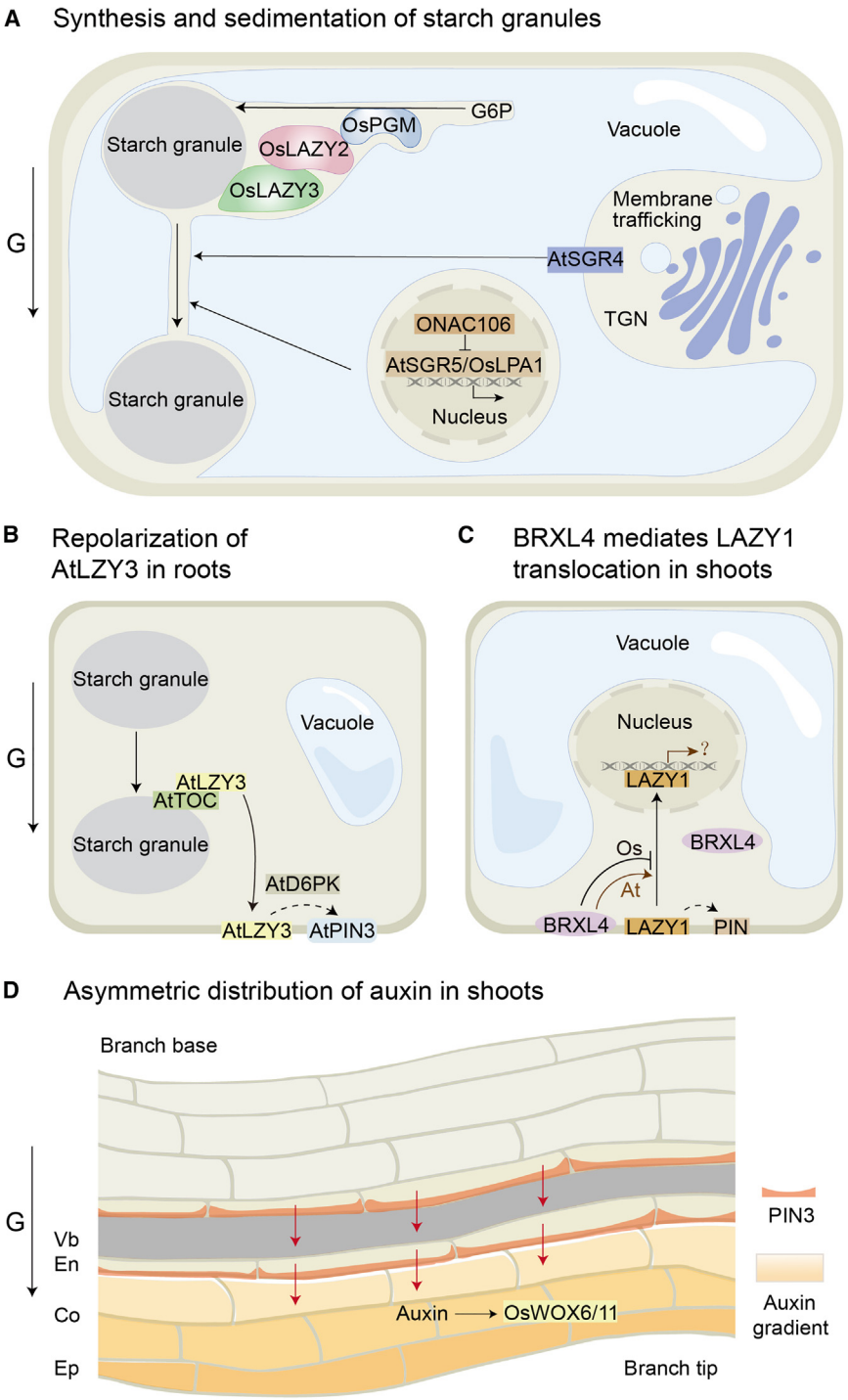

Figure 3. Gravitropism regulates shoot-tip angles.

(A) The synthesis of starch granules is regulated by OsPGM and its interactors OsLAZY2/3. Upon gravistimulation, starch granules sediment in the direction of gravity, a process modulated by ONAC106, AtSGR5/OsLPA1, and the membrane trafficking regulator AtSGR4 in shoots.

(B) AtLZY3 associates with starch granules by interacting with AtTOC and is relocalized to the near-land side of the PM. Furthermore, AtLZY3 may regulate AtPIN3 via AtD6PK in the roots.

(C) AtBRXL4 at the PM facilitates the translocation of AtLZY1 into the nucleus, thereby decreasing its proportion at the PM. AtLZY1 at the PM regulates PINs by an unknown mechanism in *Arabidopsis* shoots. By contrast, in rice, OsBRXL4 stabilizes OsLAZY1 at the PM, reducing its nuclear levels and suppressing its potential transcriptional regulation.

(D) In *Arabidopsis*, polarized PINs lead to auxin accumulation on the near-land side. Similarly, in rice, the auxin response is enhanced on the near-land side, triggering the expression of OsWOX6/11. Abbreviations: G6P, glucose-6-phosphate; OsPGM, PHOSPHOGLUCOMUTASE; AtSGR4/5, SHOOT GRAVITROPISM4/5; ONAC106, *Oryza sativa* senescence-associated NAM/ATAF1/ATAF2/CUC2 106; OsLPA1, LOOSE PLANT ARCHITECTURE1; AtLZY3, AtLZY3 (AtLAZY4); AtTOC, TRANSLOCON AT THE OUTER ENVELOPE MEMBRANE OF CHLOROPLASTS; AtD6PK, D6 PROTEIN KINASE; BRXL4, BREVIS RADIX LIKE4; OsWOX6/11, WUSCHEL-RELATED HOMEOBOX 6/11; TGN, trans-Golgi network; PM, plasma membrane; Vb, vascular bundle; En, endodermis; Co, cortex; Ep, epidermis; G, gravity.

PYROPHOSPHORYLASE (*AGPL1*), lead to impaired shoot gravitropism and increased branching angles (Okamura et al., 2013; Huang et al., 2021). Notably, two members of the rice LAZY family, OsLAZY2 and OsLAZY3, unlike other family members, act as essential cofactors of PGM in the formation of starch granules. Here, we use the format OsLAZY to represent the rice proteins and AtLZY for the *Arabidopsis* proteins. OsLAZY3 associates with starch granules through its tryptophan-rich region (TRR) domain and interacts with OsLAZY2, which in turn associates with the starch biosynthetic enzyme plastidic PGM to regulate starch biosynthesis in rice tillers (Huang et al., 2021). Both *oslazy2* and *oslazy3* mutants exhibit a complete absence of starch granules at the base of the rice leaf sheath, and the OsLAZY2–OsLAZY3 module determines starch granule formation and contributes to a reduction in rice tiller angle (Figure 3A) (Cai et al., 2023). This indicates that functional innovations within the LAZY family have evolved across various species.

Amyloplast sedimentation is comprehensively controlled during gravity perception and branching angle formation. SHOOT GRAVITROPISM5 (SGR5), a C2H2-type zinc-finger protein located in the nuclei of the shoot endodermis, modulates amyloplast sedimentation (Morita et al., 2006). A functional ortholog of *SGR5*, *Loose Plant Architecture1* (*LPA1*) in rice, also facilitates the sedimentation rate of amyloplasts, thereby reducing the angles of tillers and leaves. Furthermore, an *lpa1* mutant exhibited broader developmental phenotypes, including a thicker stem, shorter internodes, thickened cell walls, and defective vascular formation, in addition to a prostrate growth habit (Wu et al., 2013). These

broad developmental defects may be caused by its distinct downstream regulator, the auxin transporter PIN1a (Sun et al., 2019). In addition, the upstream regulator of *LPA1* is a senescence-associated NAM/ATAF1/ATAF2/CUC2 (senNAC) transcription factor, ONAC106, which represses the expression of *LPA1* and thus increases the tiller angle (Sakuraba et al., 2015). In addition, the sedimentation of amyloplasts is affected by the vacuole and its membrane trafficking. For instance, the Qb-SNARE VTI11, also known as ZIGZAG (ZIG) or SGR4, regulates membrane trafficking between the trans-Golgi network and the vacuole. Mutation of *SGR4* resulted in defective, slower sedimentation of starch granules and abnormal shoot gravitropism, leading to a pronounced increase in both branching angle and tip angle (Figure 3A) (Hashiguchi et al., 2010).

Shoot gravitropism and branching-angle formation were only partially impaired when starch granules were completely absent in mutants such as *oslazy2* and *oslazy3* (Huang et al., 2021; Cai et al., 2023). Moreover, the fern *Ceratopteris richardii* could slowly bend toward gravity while its amyloplasts were randomly dispersed (Zhang et al., 2019b). These results suggest the existence of amyloplast-independent pathways. It has been proposed that plants sense gravity mechanically through the cytoskeleton and/or mechanosensitive channels (Yoder et al., 2001; Perbal and Driss-Ecole, 2003; Leitz et al., 2009). Evidence from *Physcomitrella patens* showed that mutation in the microtubule-based cellular motor Gravitropism Group C (GTRC), a minus-end-directed KCHb kinesin, resulted in reversed protonemal gravitropism (Li et al., 2021c). Regarding actin, however, the actin depolymerizing agent Latrunculin B had no significant effect on root gravitropic bending within the first 10 min of gravistimulation; rather, it slightly enhanced the bending speed after 20 min when the curvature reached nearly 5°, as recorded by high-resolution imaging on a vertical-stage microscope (Xu et al., 2021). This observation hints that the integrity of actin filaments may not play a critical role in root gravity perception. Future studies could clarify the role of actin and microtubules in the gravity perception of branches by applying corresponding inhibitors and observing starch sedimentation rates and branch bending dynamics. The mechanosensitive channel MCA1 has also been proposed to mediate an increase in cytoplasmic Ca^{2+} upon gravistimulation in *Arabidopsis* hypocotyls (Nakano et al., 2021). However, whether and how Ca^{2+} concentrations change in statocytes and how cytoplasmic Ca^{2+} transmits information about the direction of gravity are largely unknown. Furthermore, organelles or plastids that lack starch granules but have a high density might function similarly to amyloplasts. It remains to be investigated whether LAZY colocalizes with these empty plastids in starchless mutants as a gravity-sensing mechanism independent of starch granules.

Transduction: LAZY1/4 connect amyloplast sedimentation to auxin asymmetric responses

A series of biochemical signals are generated upon gravity perception, ultimately leading to auxin asymmetry and differential cell growth. Recent studies in roots have shed light on the classic question of how gravity perception is converted into physiological responses (Chen et al., 2023b; Nishimura et al., 2023). Gravistimulation induces the phosphorylation of AtLZY3 (also referred to as AtLAZY4) and its interaction with TRANSLOCON AT THE OUTER ENVELOPE MEMBRANE OF CHLOROPLASTS (TOC) surface proteins on amyloplasts, driving the sedimentation of AtLZY3 together with the starch granules and its relocalization to the plasma membrane (PM) facing gravity (Chen et al., 2023b). The polarized AtLZY3 may further regulate the relocalization of D6 PROTEIN KINASE (D6PK), which activates PIN3, to the lower-side membrane after gravistimulation (Figure 3B) (Kulich et al., 2024). Subsequently, the PIN3-mediated redistribution and accumulation of auxin on the lower side inhibits cell elongation by alkalizing the cell wall via rapid cytoplasmic auxin signaling mediated by AFB1 and slower nuclear auxin signaling mediated by TIR1 (Li et al., 2021b; Serre et al., 2021; Chen et al., 2023a; Dubey et al., 2023). In shoots, it remains unclear how LAZYs respond to amyloplast sedimentation and modulate auxin transport and how various auxin signaling processes promote shoot cell expansion during gravitropism and the formation of branching angles.

The *LAZY* gene was first identified in rice through map-based cloning (Li et al., 2007). The *oslazy1* mutants exhibited a tiller angle of more than 50°, compared with about 10°–20° in the wild type. OsLAZY1 is localized in the PM and nucleus, and its nuclear localization is essential for determining tiller angle in rice, indicating that its main function is mediated by the transcriptional regulation of downstream genes (Li et al., 2019). Interaction assays in yeast and transient expression in tobacco leaves showed that maize LAZY1 interacts with the nucleus-localized auxin signaling co-receptor IAA17, raising the possibility that LAZY1 regulates the auxin signaling pathway for differential auxin responses (Dong et al., 2013). However, whether nuclear LAZY1 directly modulates auxin signaling *in vivo* and how its translocation responds to amyloplast sedimentation in shoots remain unknown.

Analysis of another *OsLAZY* gene, *OsLAZY4*, which encodes a C3H2C3-type RING finger E3 ligase, showed that *oslazy4* mutants exhibited a weaker prostrate phenotype than *oslazy1* and suggested that *OsLAZY4* functions upstream of *OsLAZY1*. The effects of mutations in *OsLAZY4* and *OsLAZY2* were additive, supporting the existence of two independent pathways in rice: *OsLAZY4* and the starch granule biosynthesis-regulated *OsLAZY2*–*OsLAZY3*, both of which are upstream of *OsLAZY1* (Wang et al., 2024).

Unlike their rice homologs, the *AtLZY* genes function redundantly. Only *atlzy1* single mutants displayed a significant prostrate phenotype, demonstrating that *AtLZY1* is the principal determinant of inflorescence branch angle among the four family members (Yoshihara et al., 2013; Yoshihara and Spalding, 2017). In contrast to findings in rice, it is the PM fraction of AtLZY1, rather than its nuclear fraction, that functions in shoot gravitropism and branching-angle formation in *Arabidopsis* (Yoshihara et al., 2013; Li et al., 2019). For instance, AtLZY2 expressed at the PM under the *AtLZY1* promoter can rescue the branching-angle phenotype of *atlzy1* (Yoshihara and Spalding, 2020). The functional AtLZY1 at the PM aligns with the hypothesis that AtLZY1 regulates PM-localized auxin transporters to achieve asymmetric auxin distribution.

How AtLZY1 modulates asymmetric auxin responses in shoots remains to be explored. Among the five regions of AtLZY1, regions I, II, and V are crucial for its function, and region I is required for its PM localization (Yoshihara and Spalding, 2020). In addition,

region V, the conserved C terminus, contains a small motif resembling the ERF-associated amphiphilic repression (EAR) motif, which is generally involved in recruiting proteins for transcriptional repression (Taniguchi et al., 2017). Future studies could distinguish the downstream pathways of PM- and nuclear-localized LAZY1 by manipulating its subcellular localization via PM-anchoring and nucleus accumulation, respectively, and identifying their specific interactors or through suppressor screening.

Regions I–IV of AtLZY1 also appear in TILLER ANGLE CONTROL1 (TAC1), which belongs to the IGT (GψL(A/T)IGT) family, together with *LAZY1/DEEPER ROOTING* (*DRO*) (Waite and Dardick, 2021). In *Arabidopsis*, *tac1* mutants exhibited the opposite phenotype to *atlzy1*, being profoundly erect. Furthermore, the *atlzy1* branch phenotype was epistatic to *tac1*, suggesting that *TAC1* could be upstream of *AtLZY1* (Hollender et al., 2020). However, *Arabidopsis* mutants of both genes did not share notable overlapping genes in a transcriptome analysis, suggesting that distinct pathways may be mediated by *AtLZY1* and *TAC1* (Hollender et al., 2020). Similarly, in rice, *TAC1* and *TAC3* function oppositely to *OsLAZY1* (Yu et al., 2007; Dong et al., 2016). By contrast, *TAC4* acts similarly to *OsLAZY1*, as *tac4* mutants are prostrate like *oslazy1.* In particular, *TAC4* facilitates the expression of *YUCCAs* to increase auxin levels and eventually promote asymmetric distribution of auxin in rice (Li et al., 2021a). IGT family members thus regulate gravitropism and branching angles in distinct pathways across species.

Another insight into the mechanism of LAZY1 comes from its interacting protein. BREVIS RADIX LIKE 4 (BRXL4) was identified in rice as interacting with the C terminus of OsLAZY1 at the PM where they are colocalized (Li et al., 2019). Moreover, BRXL4 stabilized OsLAZY1 at the PM and reduced its proportion in the nucleus, as shown by transient-expression assays. Because nuclear OsLAZY1 is the functional form, BRXL4 counteracts OsLAZY1 activity, as supported by the phenotypes of RNAi and overexpression lines (Li et al., 2019). Similarly, *brxl4* mutants in *Arabidopsis* exhibit upright branches, contrasting with those of *atlzy1*. Interestingly, BRXL4 promotes the removal of AtLZY1 from the PM where it actually functions, thereby again counteracting the function of AtLZY1 (Che et al., 2023). It remains unclear how rice BRXL4 stabilizes OsLAZY1 at the PM but *Arabidopsis* BRXL4 causes AtLZY1 to detach from the PM. Whether and how PM-localized AtLZY1 regulates auxin transport in *Arabidopsis* and whether and how nuclear OsLAZY1 regulates downstream gene expression to establish auxin asymmetric responses in rice also remain to be determined (Figure 3C and 3D).

Anti-gravitropic offset: A counteracting mechanism behind the veil

What would the branching angle be if gravity were removed? Microgravity simulators like clinostats reduce the effect of gravity by rotating plants and disabling their perception of gravity. Clinorotation causes plant shoots to grow downward rather than in a random direction. This indicates the existence of a mechanism specifically directing shoots to grow downward, in contrast to negative gravitropism, which causes shoots to grow upward. Such a mechanism has been proposed as the AGO, which counteracts gravitropism and maintains the GSA in shoots and roots (Roychoudhry et al., 2013; Roychoudhry and Kepinski, 2015).

A natural location with the least effect of gravity is the International Space Station. However, plants in space experiments have usually exhibited upright growth, as lights were positioned above them. It is unclear whether branches could bend specifically downward owing to AGO or whether they would grow in all directions in space. Data from reduced upward-growing mutants on Earth may provide some clues. For instance, *Arabidopsis atlzy1,2,3* mutants showed very few instances of complete downward growth and nearly equal probabilities for reduced upward angles (Taniguchi et al., 2017). Strikingly, *Physcomitrella patens* with a mutation in the *GTRC* locus, which encodes a processive minus-end-directed KCHb kinesin, exhibited a complete shift to downward-growing protonemas, in contrast to the upward-growing wild type. Other mutants with protonemas that spread randomly in all directions were also observed in the same screen (Li et al., 2021c). In these experiments, mutations could result in growth in an opposite direction or random directions, leading to two hypotheses. One is that a component in the shoot gravitropism machinery changes its function, mediating both upward and downward bending. For instance, PIN3/4/7 in the branch, which normally respond to gravistimulation and are localized toward gravity, could have their polarity directed away from gravity, resulting in an opposite response. However, questions may arise regarding what signals and mechanisms are responsible for the localization of PIN proteins on the side opposite to gravity, as nothing appears to sediment away from gravitational forces. The other hypothesis is that shoot gravitropism promotes upward branch growth, whereas AGO induces downward growth. Removal of one pathway reveals the effect of the other, and deletion of a common component in both pathways leads to random growth directions. These two possibilities are not mutually exclusive, as the latter could also include common components that regulate upward and downward bending.

To explore the effect and mechanisms of AGO, clinorotation was performed to reduce gravitropism and reveal the effect of AGO in *Arabidopsis* axillary branches (Roychoudhry et al., 2013). The auxin signaling mutants *tir1* and *auxin response factor7,19* (*arf7,19*) exhibited more changes in GSA after clinorotation, indicating a negative role of *TIR1* and ARF7/19 in the regulation of AGO (Roychoudhry et al., 2013). Because *TIR1* and *ARF7/19* positively mediate gravitropism in hypocotyls (Wang et al., 2020), the same *TIR1–ARF7/19* module appears to contribute to gravitropism while repressing AGO in shoots. On the other hand, *arf10,16* double mutants and *arf10,16,axr3* triple mutants all displayed a reduced AGO response, suggesting that *ARF10/16* and *AXR3* are positive regulators of AGO (Roychoudhry et al., 2013). Clues from hypocotyls suggest that auxin-induced *ARF10/16* inhibit cell elongation (Dai et al., 2021), and it appears that the ARF10/16 module potentially inhibits shoot gravitropism but mediates AGO by an unknown mechanism. Furthermore, the auxin transport inhibitor NPA significantly reduced changes in GSA after clinorotation (Roychoudhry et al., 2013). This suggests that auxin transport may positively mediate AGO, resembling the positive role of auxin transport

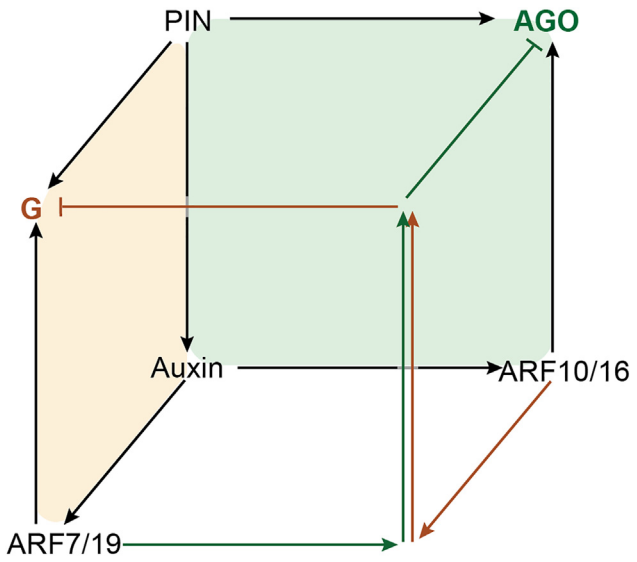

Figure 4. Auxin regulation of gravitropism and anti-gravitropic offset during tip-angle formation in *Arabidopsis*.
PINs mediate the transport and local accumulation of auxin. Upon auxin perception, the transcription factors ARF7/19 mediate auxin-induced cell expansion, resulting in the upward bending characteristic of negative gravitropism (G). In addition, ARF7/19 inhibit the anti-gravitropic offset (AGO) through unknown mechanisms. By contrast, the transcription factors ARF10/16 contribute to the AGO through unknown mechanisms, leading to downward bending. Furthermore, auxin-induced ARF10/16 suppress hypocotyl cell elongation, potentially inhibiting G. Finally, PINs mediate G, and auxin transport contributes to AGO by unknown mechanisms. Yellow lines and shading, regulators for G; green lines and shading, regulators for AGO. Abbreviations: G, gravitropism; AGO, anti-gravitropic offset; ARF7/19, AUXIN RESPONSE FACTOR7/19; ARF10/16, AUXIN RESPONSE FACTOR10/16.

during gravitropism. These findings indicate that two pairs of auxin signaling modules play opposite roles, and auxin transport may function similarly in the counteracting responses of gravitropism and AGO (Figure 4).

The above descriptions highlight the dual roles of auxin in gravitropism and AGO; however, how auxin specifically regulates branching angles remains to be determined. To provide a comprehensive view, we summarize the angle phenotypes associated with auxin mutants or related treatments. Exogenous application of auxin to decapitated tips led to an increase in both branching angle and tip angle in *Arabidopsis* and pea (Roychoudhry et al., 2013). This suggests that apical auxin remotely promotes larger branching angles. On the other hand, the auxin biosynthesis mutant *wei8,tar2* and the connective auxin transport (CAT) mutant *pin3,4,7* displayed larger branching angles than Col-0 in *Arabidopsis* (Roychoudhry et al., 2013; van Rongen et al., 2019). Accordingly, the dominant negative allele of auxin biosynthesis, *yucca1-1d*, led to reduced branching angles (Roychoudhry et al., 2013). In terms of auxin signaling, the outcomes differed: *tir1* and *arf7,19* mutants exhibited prostrate growth, whereas *arf10,16,axr3* grew more upright (Roychoudhry et al., 2013). To date, it remains largely unknown how AGO is initiated or sensed, how auxin is transported under the AGO mechanism, how the mechanisms of gravitropism and AGO differ from each other, and how they may participate in crosstalk.

Light: Communication with the crowds throughout the day

Most plants photosynthesize during the day and respire and grow faster at night (Hilty et al., 2021). Light prompts plants to stretch out with larger branch and tip angles in *Arabidopsis*, thereby increasing the photosynthetic area. By contrast, darkness and low red/far-red light conditions result in a reduction of branching angles (Roychoudhry et al., 2017), and such vertical growth provides plants with a less responsive stature for gravitropism. Thus, mutations in plant *PHYTOCHROME B* (*PHYB*) photoreceptors and overexpression of *PHYTOCHROME INTERACTING FACTOR-LIKE15* (*PIL15*) in rice led to a smaller tiller angle (Xie et al., 2019). On the other hand, light and shade vary dynamically, not only over time but also across different positions within a plant or population. Crop studies have reported that the shade-avoidance response modulates the distribution of leaf angles along the canopy, resulting in a smart canopy and increased yield for dense planting (Tian et al., 2019, 2024; Sellaro et al., 2024; Zhou et al., 2024).

Light- or dark-induced changes in branching angles are mediated by the previously described IGT family. *TAC1* mediates the light-induced increase in branching angles. Mutations in *TAC1* resulted in smaller branching and tip angles under light conditions. Notably, blue light enhanced the expression of *TAC1*, whereas far-red light and darkness suppressed it. Consequently, mutations in blue light receptors, such as *CRYPTOCHROME1/2* (*CRY1/2*) or *PHOTOTROPIN1/2* (*PHOT1/2*), or *CONSTITUTIVE PHOTOMORPHOGENESIS1* (*COP1*), which integrates multiple light signals, all led to a reduction in *TAC1* levels. In addition, Col-0 and *tac1* mutants remained erect after prolonged darkness lasting 72 h. These findings support the notion that *TAC1* integrates light signals to regulate branching angles (Waite and Dardick, 2018).

Compared with *TAC1*, *AtLZY/DRO*s are required to reduce the tip angle in response to darkness as well as for tip angle formation in the light. The tip angles of the *atlzy1,dro1,3* triple mutant and the *atlzy1,dro1,2,3* quadruple mutant remain unchanged under darkness, in contrast to the branch closure observed in Col-0. Darkness has been reported to promote *LAZY1* expression in maize (Dong et al., 2013). These findings suggest that *AtLZY1/DROs* redundantly contribute to the dark-induced reduction in tip angle. Circadian rhythms in the expression of *AtLZY/DRO* were observed under light conditions. Moreover, red and blue light induced changes in the expression of different *AtLZY/DROs*, depending on the respective photoreceptors (Waite and Dardick, 2024). However, it remains unclear how LAZY/DROs and TAC1 respond to light signaling in shoots. Some clues have been reported in studies of hypocotyls and roots, which indicated that the light signaling components PHYTOCHROME INTERACTING FACTORs (PIFs) and ELONGATED HYPOCOTYL5 (HY5) can directly bind to and positively regulate the expression of *AtLZY3*. Light caused the degradation of PIFs and reduced the expression of *AtLZY3* in hypocotyls; conversely, light promoted the accumulation of HY5, thereby enhancing the expression of *AtLZY3* in roots (Yang et al., 2020).

Insights from other hormones, key proteins, and miRNAs

In addition to describing the roles of auxin-mediated gravitropism, AGO, and phototropism, we also summarize the roles of two additional hormones—strigolactones (SLs) and brassinosteroids (BRs)—along with key proteins and miRNAs that participate in the regulation of branching angles. SLs not only inhibit shoot branching but also produce a prostrate phenotype with large branching angles. Because of the crucial role of auxin in branching and branching-angle formation, the relationship between SLs and auxin transport in the regulation of shoot branching and angles has been studied. Mutations in the SL receptor *DWARF14* (*D14*) or the signaling components *MORE AXILLARY BRANCHING2/4* (*MAX2/4*) resulted in characteristic phenotypes such as increased branching numbers and acute branching angles. On the other hand, the *pin3,4,7* triple mutants showed a normal branching number but obtuse branching angles. Additional mutations in PIN3/4/7 were able to partially suppress the acute branching angles of the SL mutants (van Rongen et al., 2019). SL biosynthetic or signaling-related mutants were identified in a suppressor screen for *oslazy1.* Unlike OsLAZY1, which regulates tiller angle via auxin transport, SLs are believed to regulate rice gravitropism and tiller angle via auxin biosynthesis (Sang et al., 2014). Furthermore, BRs also influence tiller angle. For instance, the *dwarf2* (*d2*) mutants, which are defective in a BR biosynthetic enzyme, exhibit a reduced tiller angle (Dong et al., 2016). BRs activate the expression of *LEAF AND TILLER ANGLE INCREASED CONTROLLER* (*LIC*), which encodes a CCCH-type zinc-finger protein, leading to an erect phenotype in rice (Wang et al., 2008; Zhang et al., 2012).

Many transcription factors and key proteins that regulate branching angles in crops have been discovered through forward genetics studies (Wang et al., 2022). These can be classified into four categories based on the target of their effects: auxin homeostasis, auxin transport, auxin signaling, and cell expansion and division (Supplemental Table 1). For auxin homeostasis, *BASIC LEUCINE ZIPPER49* (*bZIP49*), which encodes a TGA-class transcription factor, directly promotes the expression of the indole-3-acetic acid-amido synthetase *GRETCHEN HAGEN 3* (*GH3*), leading to a reduction in free auxin levels (Ding et al., 2021). Consequently, overexpression of *bZIP49* leads to a prostrate phenotype, whereas rice *bzip49* mutants display a compact architecture.

For auxin transport regulation, the zinc-finger transcription factor *PROSTRATE GROWTH 1* (*PROG1*), which is expressed predominantly in the rice axillary meristem, directly binds to two promoter regions of *OsLAZY1* and represses its expression, thus regulating auxin transport (Wang et al., 2023a). Substitution of an amino acid in PROG1 has been selected during rice domestication (Jin et al., 2008; Tan et al., 2008). Through high-resolution transcriptomic analysis in rice, *HEAT STRESS TRANSCRIPTION FACTOR2D* (*HSFA2D*) was identified as being transiently upregulated in the early gravitropic response, promoting the expression of *OsLAZY1*. The *hsfa2d* mutants exhibit a less severe phenotype than the prostrate phenotype of *oslazy1* in rice, and their double mutants resemble *oslazy1*, supporting the notion that *HSFA2D* is a positive upstream regulator of *OsLAZY1*, which subsequently regulates auxin transport (Zhang et al., 2018). In peach trees, the *WEEP* gene was identified as regulating the weeping phenotype (Hollender et al., 2018). Starch granules sedimented normally in *weep mutants,* but auxin asymmetry was affected, indicating that *WEEP* may regulate branching angles by modulating auxin transport (Kohler et al., 2024).

For the modulation of auxin signaling, the transcription factor GROWTH-REGULATING FACTOR7 (GRF7) directly enhances the expression of *ARF12*, thereby influencing the auxin signaling pathway and reducing the tiller angle in rice. GRF7 also directly binds to and promotes the expression of the cytochrome P450 gene *CYP714B1*, which affects gibberellic acid (GA) synthesis and contributes to the regulation of tiller angle (Chen et al., 2020). In addition, overexpression of miR167a in rice inhibits *ARF12/17/25* to suppress auxin asymmetric responses, subsequently reducing the expression of *WOX6/11* and enlarging the tiller angle (Li et al., 2020).

The downstream events of auxin signaling include cell expansion and cell division. Some TCP transcription factors that affect branching angles can directly regulate cell elongation or proliferation. For instance, *TILLER INCLINED GROWTH1* (*TIG1*), which is expressed primarily at the far-land or adaxial side of the tiller base, promotes the expression of *EXPANSIN A3* (*EXPA3*), *EXPB5*, and *SMALL AUXIN-UP RNA39* (*SAUR39*), thereby facilitating cell elongation and enlarging the tiller angle in wild rice. By contrast, *indica* cultivars harbor variations in the *TIG1* promoter, displaying reduced *TIG1* expression and erect tiller growth (Zhang et al., 2019a). Genetic analysis showed that double mutations of *TIG1* and *PROG1* or *TAC1* led to enhanced erectness in tillers of nearly 5°–10°. Thus, their functions in the regulation of tiller angle are additive (Zhang et al., 2019a). On the other hand, near-ground auxin accumulation in rice induced the expression of *WUSCHEL-RELATED HOMEOBOX6* (*WOX6*) and *WOX11*, which are genetically downstream of *OsLAZY1* (Zhang et al., 2018). The *wox6,11* mutant shows significantly larger tiller angles and a prostrate phenotype (Zhang et al., 2018). In addition, WOX11 interacts with H3K27me3 demethylase to activate the gene expression required for cell proliferation during rice shoot development (Cheng et al., 2018).

Overall, most regulators of branching angles discovered to date operate through auxin biosynthesis, transport, or signaling or the downstream processes of cell expansion and division (Supplemental Table 1). These findings not only emphasize the crucial role of auxin but also reveal a variety of specific pathways through which auxin regulates branching angles. In addition to auxin, other hormones such as SLs and BRs also contribute to the formation of branching angle.

ANGLE DYNAMICS: CHALLENGING ASPECTS OF PLANT ARCHITECTURE AND ADAPTATION

Branching angles change during branch development. Branching angle often increases after initial branch formation, enabling expansion of the assimilation area and creating space for young branches (Mantilla-Perez and Salas Fernandez, 2017). However, the branching angle can also decrease after the initial spreading phase. The tiller angle in the IL55 rice cultivar primarily increased

to about 17° at 60 days after sowing (DAS) and then decreased to 0° at 100 DAS at the heading stage, a change that is believed to help avoid shading in crowded fields (Yu et al., 2007). For grass and woody plants with comprehensive architecture, typical upward or downward curvatures are formed during branch development (Figure 2C, 2D, and 2F–2I). Despite extensive research on the mechanisms of bud outgrowth and elongation, it remains unclear whether hormones, sugars, and environmental stimuli regulate branching angles during bud release and elongation (Bertheloot et al., 2020; Kerr et al., 2021; Yu et al., 2021; Cao et al., 2023; Zuccarelli et al., 2023; Song et al., 2024).

The control of branching angle throughout development has been less studied. As a branch develops, its weight, thickness, and age all increase. How gravitropism, AGO, phototropism, and other regulators coordinate with the weight and thickness of a branch to dynamically regulate branching angle during development remains to be explored. A recent study revealed the molecular mechanism by which branch weight modulates radial growth and, consequently, branch thickness. Upon application of weight to *Arabidopsis*, the PIN3 domain from the lower side of the cell was expanded to the lateral side, leading to radial expansion of auxin accumulation and initiation of cambium activity (Carrió-Seguí et al., 2024). Notably, the translocation of PINs is also involved in gravitropism, AGO, and phototropism (Ding et al., 2011), suggesting that it might be the central hub of branching-angle regulation during development.

Aside from branch development itself, branching angles are also affected by positional cues within the entire plant. When a tree falls and the trunk generates new shoots, these new shoots grow vertically. In addition, decapitation can induce a decrease in the branching angle of lateral branches, which can take the place of the removed leader in *Arabidopsis* and pea (Roychoudhry et al., 2013). In these cases, the identity of the branch, whether it is a side branch inhibited by apical dominance or apical auxin or it will become the leading branch, can solely determine its branching angles. This reinforces the point that auxin serves as a key signal by which the shoot apex remotely regulates the axillary branches and their branching angles. On the other hand, branches may possibly affect the angle of the main stem. For instance, the main stem of *atlzy1* can be inclined to the opposite side of the axillary branches, resulting in a zig-zag shape, in contrast to the straight main stem observed in the wild type under certain growth conditions (Taniguchi et al., 2017; Yoshihara and Spalding, 2020). This resembles the sympodial inflorescence observed in tomato, in which initiation of the floral meristem pushes away the sympodial inflorescence meristem (Lippman et al., 2008). These observations suggest that the interaction between axillary branches and the main stem regulates their respective angles. Future studies could explore the regulatory network that connects the branches and the main stem.

As one of the key factors that contribute to plant architecture and its adaptation to the environment, branching angle is correlated with other traits, including branching number, plant height, and disease resistance. For instance, mutants with reduced tiller angles, such as *phyb1,2* and *liguleless1* (*lg1*), display increased plant height (Shi et al., 2024). *Plant architecture and yield 1* (*pay1*) mutants exhibit smaller tiller angles, lower tiller numbers, and greater plant height (Zhao et al., 2015). Overexpression of *DEHYDRATION RESPONSE ELEMENT BINDING PROTEIN 1B* (*DREB1B*) led to a reduced branching angle and a significant decrease in plant height and branch length in cotton (Ji et al., 2021). Moreover, LPA1 and PIN1a not only regulate the tiller angle but also confer resistance to rice sheath blight disease through the induction of pathogen-resistance genes, coupling tiller angle and defense-related gene activation (Sun et al., 2019). Future studies could analyze the crosstalk among branching angles, branching number, stress resistance, and crop yield during the dynamic regulation of branching angle.

In summary, branching angles dynamically change at various stages of plant development, balancing leaf-area expansion with shade avoidance. Despite the significance of branching angle, the mechanisms that regulate its dynamics remain largely unknown. Evidence indicates that auxin transport primarily orchestrates the developmental and environmental regulation of branching-angle formation. However, how gravitropism, AGO, and phototropism change and coordinate with increasing branch weight, thickness, and age; how branches and the main stem communicate with each other, possibly via auxin signaling; and how branching angles coordinate the regulation of branching number, stress resistance, and crop yield are exciting open areas that await further exploration.

EVOLUTION OF PLANT BRANCHING ANGLES

Branching angles, together with branching type, have evolved multiple times throughout plant history. The evolutionary analysis of branching forms has been the subject of a recent systematic review (Harrison, 2017); here, we describe the regulatory mechanisms of branching angles in algae, non-vascular plants, early diverging vascular plants, and seed plants. Because plant architecture varies across many lineages, the mechanisms that govern angle formation appear to have evolved repeatedly according to the shape of plants.

Among various algae, some species exhibit particularly complex architecture. For example, the red alga *Chondrus crispus* has developed a unique multicellular organization within the Rhodophyta. Attached to rocks in the lower intertidal and shallow subtidal zones, *C. crispus* repeatedly branches dichotomously at acute angles, forming a broom-like architecture and increasing its photosynthetic area (Bourgougnon, 2014) (Figure 5A). Within the Charophyta—a group of green algae closely related to land plants—*Chara braunii* demonstrates a distinctive architecture characterized by whorled branchlets that harbor sexual structures at their axils (Nishiyama et al., 2018). The angle between the young branchlet cells and the stem is initially small and then increases to a perpendicular angle in mature branchlets, resembling the dynamics of branching angles in vascular plants (Figure 5B). The branching angle of *Chara* may be regulated by the nodal cells to which the branchlet cells are attached during development, providing space for new whorls of branchlets.

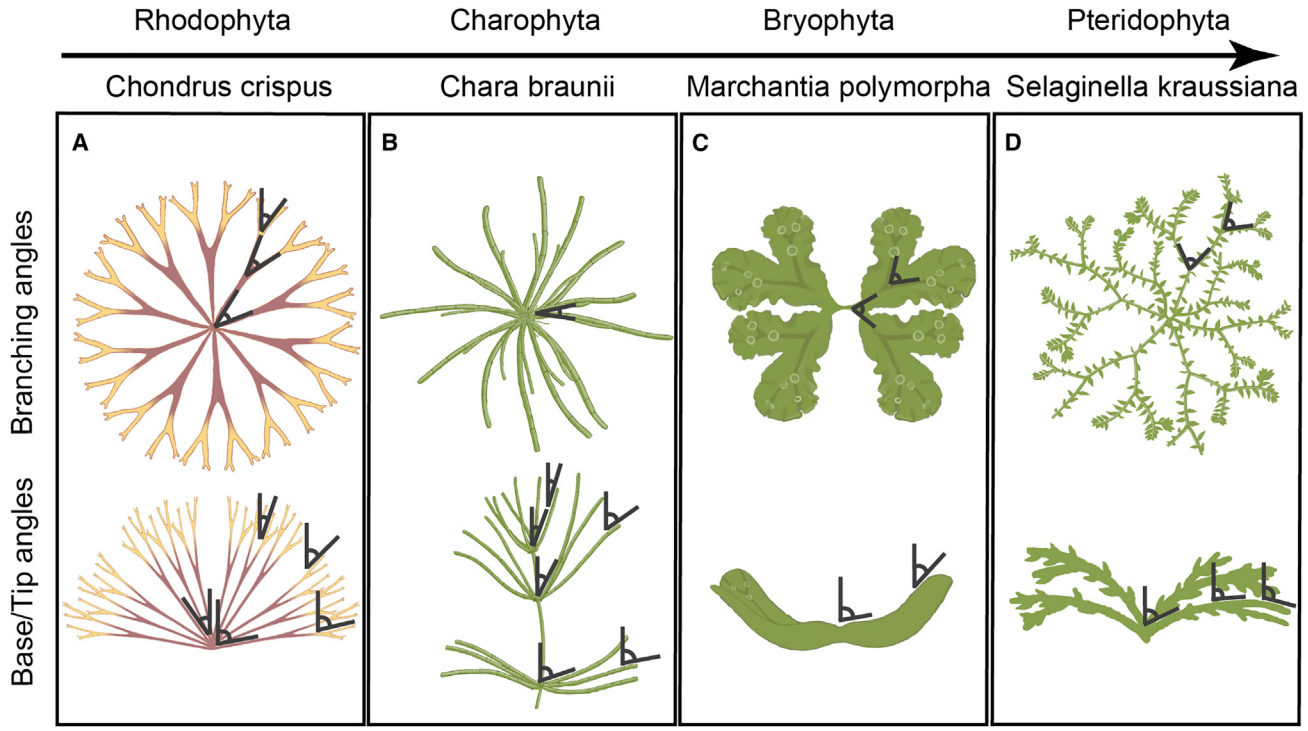

Figure 5. Conserved branching angles during plant evolution.

(A) In Rhodophyta, *Chondrus crispus* attaches to rocks and forms a unique architecture through dichotomous branching with acute angles between axillary branches. Its base and tip angles range from acute to right angles.

(B) In Charophyta, *Chara braunii* exhibits a notable three-dimensional form. The angle between branchlets is acute. The base and tip angles of the whorled branchlets are initially acute but develop into right angles later.

(C) In Bryophyta, the liverwort *Marchantia polymorpha* branches dichotomously at acute angles. The base angle of the thallus is nearly right, and its tip angle is acute.

(D) In Pteridophyta, the lycophyte *Selaginella kraussiana* exhibits a sprawling, prostrate growth habit characterized by anisotomous dichotomy with acute angles between axillary branches. The base and tip angles dynamically range from acute to obtuse. The upper panels provide an overview, showing branching angle, and the lower panels show a side view of the plants, indicating the base and tip angles.

Plant terrestrialization was accompanied by significant changes in body plan and morphological complexity, facilitating adaptation to various land habitats. Two non-vascular plants with distinct architectures have been studied to understand the molecular mechanisms that underlie shoot gravitropism and branching-angle formation. In the liverwort *Marchantia polymorpha*, the thallus grows indeterminately with repeated dichotomous branching (Wang et al., 2023b; Streubel et al., 2023). The angle between branches is again acute, enabling more branches to cover the horizontal space (Figure 5C). The vegetative offspring from the thallus produces gemmae. Four-day-old gemmalings exhibit two shoots at angles of about 29°–38° to the ground (Figure 5C), whereas the shoots of *pin1-1* mutants are more prostrate, with only a 4° angle to the ground. Similarly, overexpression of *PINZ* leads to erect branching of the thallus (Tang et al., 2024). These findings suggest that auxin transporters modulate the tip angle of *Marchantia* similarly to those in vascular plants (Fisher et al., 2023). On the other hand, transient treatment with the CLAVATA3 (CLV3) peptide or its expression driven by the promoter of the auxin biosynthesis gene *YUCCA2* at apical notches leads to an increase in stem cells, multichotomous branching, and a decrease in the angle between two branches (Hirakawa et al., 2020). These findings demonstrate that auxin transport regulates the base and tip angles of *Marchantia*, whereas auxin biosynthesis is possibly involved in the CLV3-mediated stem cell formation that regulates the angle between branches, or the branching angle. These findings support the idea that the branching angle is initially established by stem cells, whereas auxin-transport-mediated gravitropism controls the angles with respect to gravity. In contrast to *Marchantia*, the moss *Physcomitrium patens* features a colonial structure consisting of elongating, branching horizontal protonema filaments and outgrowing upward gametophores. The length and growth direction of the protonema determine the size of the colony, whereas the density, position, and angle of the gametophores primarily contribute to the colony's vertical shape. Studies have shown that kinesin regulates the direction of protonema filaments, and *AFB* auxin receptors and auxin response factors in clade D facilitate protonemal branching and promote the initiation of gametophores (Johri and Desai, 1973; Prigge et al., 2010; Bascom et al., 2023), thus shaping the moss colony.

An early diverging vascular plant, the lycophyte *Selaginella kraussiana*, exhibits a sprawling prostrate growth habit characterized by anisotomous dichotomy, specifically, a dominant branch overtopping a minor branch (Spencer et al., 2023). The alternation of major and minor branches on the left and right produces a zigzag shoot architecture with acute branching angles (Figure 5D). In

addition, the tip angle increases during development, contributing to the prostrate phenotype. Studies have shown that auxin from the branch tip, transported by canonical PINs in the vasculature, modulates the number of dichotomies and angled branches but has no notable effect on the length and angles between branches. This suggests that auxin transport may not affect the branching angle established by the meristems (Spencer et al., 2023).

Seed plants display a diverse array of architectures and branching angles. In gymnosperms, studies have shown that approximately 90% of *Picea abies* have branching angles less than 70°, with approximately 60% of these angles ranging between 40° and 70° (Fabrika et al., 2019). By contrast, *Ginkgo biloba* can develop into either a pillar shape with acute branching angles or a spreading form with more obtuse angles, often correlated with the tree's sex. The mechanisms that regulate branching angles in gymnosperms are far from well understood. However, compression wood has been observed to develop on the lower side of leaning branches in many gymnosperms, pushing the stem upright during gravitropism. By contrast, tension wood, characterized by higher cellulose content, forms on the upper side of leaning woody branches in many angiosperms, acting by pulling the stem upright (Groover, 2016). Studies in *Populus* trees demonstrated that initial, rapid bending occurred at the growing shoot tip within the first day after a 90° gravistimulation, followed by gradual bending of the entire shoot over 2 weeks. Time-series transcriptome analysis revealed that transcriptional regulation peaked at 48 h. Specifically, hormone regulation, including that of auxin, GA, BR, ethylene, abscisic acid, jasmonic acid, and salicylic acid, was observed at 24 h in the lower-side wood, which is commonly referred to as opposite wood in angiosperms. By contrast, changes in genes associated with saccharides, development, and epigenetic processes occurred at 24–48 h in tension wood (Zinkgraf et al., 2018). Notably, GA promotes gravistimulated bending by facilitating the maturation of gelatinous layers or G-layers and the formation of tension wood through the Class I KNOX homeodomain transcription factor gene ARBORKNOX2 (ARK2) in 2 weeks. In addition, ARK2 induces polarization of the auxin transporter PIN3 toward gravity in 4 days, which may trigger the formation of tension wood on the upper side and opposite wood on the lower side of the gravistimulated *Populus* stem (Gerttula et al., 2015).

Overall, branching angles have repeatedly evolved alongside branching forms. The branching angle between shoots is established by interactions between shoot meristems that are regulated by stem-cell regulators such as CLV3, as seen in the acute branching angle of dichotomous branching in the red alga *C. crispus*, the liverwort *M. polymorpha*, and the lycophyte *S. kraussiana*. By contrast, the base and tip angles with respect to gravity are regulated by gravitropism and its regulators involving auxin transport PINs, as seen in the liverwort *M. polymorpha* and in seed plants. PIN may have begun regulating the tip angle early in plant terrestrialization, as studies have shown that the PIN from the green alga *Klebsormidium flaccidum* cannot complement the severely defective shoot development of *Arabidopsis pin1,3,4,7* quadruple mutants; however, these defects can be substantially rescued by PINs from the land plants *M. polymorpha* and *Physcomitrium patens* (Zhang et al., 2020). Furthermore, gymnosperms and woody angiosperms exhibit slow gravitropism, mediated by the induction of compression wood to provide a pushing force and tension wood to generate a pulling force, respectively. This process requires extensive development of the cell wall.

CONCLUDING REMARKS AND PERSPECTIVES

The coordination of branching angles with branching types results in diverse plant shapes that can adapt to varied environments. Branching angles include the branching angle, the base angle, and the tip angle. The branching angle between the shoots is possibly initiated by inhibitory mechanisms between meristems, such as the angle between two dichotomous branches in *Marchantia* and the angle between the flower and the inflorescence in tomato; this may also apply to the angle between branches and the main stem in vascular plants. Moreover, the branching angle can also vary during branch development in vascular plants. By contrast, the base and tip angles relative to the direction of gravity have received intensive study. The primary mechanism that universally regulates base and tip angles in dicots and monocots is shoot negative gravitropism. Mutants of genes related to starch granule biosynthesis or sedimentation and to auxin transport or signaling have all shown significant changes in branching angles (Supplemental Table 1). On the other hand, an AGO has also been proposed that would allow branches to grow downward, counteracting gravitropism and thus maintaining a characteristic gravity-setpoint angle. The molecular mechanism of the AGO is highly intertwined with that of gravitropism in auxin transport and signaling, but little is known about how it is perceived and how it functions distinctively from gravitropism (Figure 4). Identified regulators of base and tip angles can be categorized into those involved in auxin biosynthesis, transport, and signaling, as well as the downstream processes of cell expansion and cell division (Supplemental Table 1). The PIN auxin transporter modulates tip angle in the liverwort *Marchantia*, highlighting the conserved and comprehensive roles of auxin in regulating base and tip angles. On the other hand, a significant feature of the contribution of branching angles to plant architecture is their dynamics, which are currently far from well understood. Future research on how gravitropism, AGO, phototropism, and other regulators coordinate with the weight and thickness of the branch to regulate branching angle during development will shed light on the mechanism of angle dynamics. These findings will contribute to a comprehensive understanding of dynamic shoot development and the formation of plant architecture.

Branching angles significantly influence crop yield by shaping plant architecture. Steeper branch or leaf angles in the canopy enhance light penetration and CO_2 uptake in the lower canopy layers, thus improving photosynthetic efficiency and increasing crop yield (Burgess et al., 2017; Wang et al., 2022; Basu and Parida, 2023). Numerous case studies in various crops have demonstrated that the modification of branch or leaf angles can produce substantial yield improvements. For instance, a mutation of *LEAF ANGLE ARCHITECTURE OF SMART CANOPY1* (*LAC1*) in maize produced a "smart canopy" with reduced angles in the upper canopy, enhancing photosynthesis under

dense planting conditions. At a planting density of 135 × 10^3 plants ha^{-1}, the *lac1* mutants yielded an additional 5 g per plant and 0.5 tons per hectare compared with the wild type. However, the *lac1* mutants exhibited lower yields when planted at densities below 67.5 × 10^3 plants ha^{-1} (Tian et al., 2024). Similarly, the brassinosteroid-deficient *dwarf4-1* rice mutants, which displayed a reduced leaf angle, achieved notable grain yields in dense plantings, even without additional fertilizer (Sakamoto et al., 2006). Rice varieties with erectophile canopies yielded 13% more and produced 24% more grains per unit area, although they had a 9% lower grain weight compared with planophile lines (Richards et al., 2019). Moreover, overexpressing *LPA1* in rice not only reduced the tiller angle but also enhanced resistance to sheath blight disease (Sun et al., 2019). Thus, the identification of genes that specifically regulate the angles of branches or leaves is a promising strategy for optimizing crop architecture to maximize yields in dense planting environments. Future efforts could focus on integrating gene discovery with advanced breeding techniques that leverage unmanned aerial vehicles for large-scale phenotyping, artificial intelligence-based modeling for phenotype analysis, and pan-genome analysis of wild germplasm and landraces (Imam et al., 2016; van Eeuwijk et al., 2019; De Swaef et al., 2021; Basu and Parida, 2023). The CRISPR-Cas system provides an efficient tool for targeted gene editing and can significantly expedite the development of crops with ideal architecture and enhanced yields.

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AUTHOR CONTRIBUTIONS

L.L. wrote the manuscript. C.Y. and W.M. created the figures, assisted with the literature research, and revised the table and manuscript. J.F. summarized the table and helped with revision.

SUPPLEMENTAL INFORMATION

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